ROSHNI SINGH

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EDUCATION

Vellore Institute of Technology B.Tech – Computer Science and Engineering	2022 – 2026	8.2
Delhi Public School Bangalore South Senior Secondary Education (12th)	2022	74
Delhi Public School Bangalore South Secondary School Examination (Class X)	2020	90

TECHNICAL SKILLS

Languages: Java, Python, JavaScript
Databases: MongoDB, SQL
Frameworks & Libraries: React.js, FastAPI, PyTorch, TensorFlow, Pandas, NumPy
AI / ML: CNN, Transformers, Deep Learning, Reinforcement Learning, PSD Analysis
Tools: Git, GitHub, SUMO, HuggingFace, Streamlit
GenAI: RAG, LangChain, ChromaDB, Ollama, Prompt Engineering

EXPERIENCE

Mathological Technologies Private Limited– AI/ML Engineering Intern

- Working on an AI-driven Predictive Maintenance System for Metro Rail assets using fault-event and IoT sensor data.
- Performing exploratory data analysis (EDA), data validation, and visualization to identify patterns, trends, and potential failure indicators in operational datasets.
- Assisting in data preprocessing, cleaning, and feature engineering to transform raw sensor and maintenance data into machine learning-ready datasets.
- Researching predictive maintenance methodologies, failure prediction techniques, and machine learning approaches for rail asset monitoring.
- Collaborating with the team to support the development of data-driven solutions aimed at improving asset reliability, reducing unplanned breakdowns, and optimizing maintenance planning.

PROJECTS

Seismic Event Detection System GitHub ↗
Stack: PyTorch, CNN, Transformers, STEAD Dataset, Signal Processing
- Developed a deep learning system using CNN and Transformer architectures for earthquake detection, magnitude estimation, and epicenter location prediction.
- Integrated Power Spectral Density (PSD) features for enhanced frequency-domain analysis, improving magnitude prediction accuracy.
- Built a multi-stage pipeline on STEAD seismic waveform data for event classification, geolocation, and risk-based alert generation.
- Designed evaluation, testing, and visualization modules with an interactive dashboard for seismic monitoring and prediction analysis.

InsightPDF – RAG Document Intelligence Platform GitHub ↗
Stack: LangChain, ChromaDB, HuggingFace, Streamlit, Ollama (Llama-3)
- Built a Streamlit-based RAG app that ingests large PDFs and provides structured summaries and conversational Q&A over document content.
- Implemented a modular LangChain backend for document classification, summarization, retrieval, and conversational agents.
- Integrated ChromaDB and HuggingFace embeddings for persistent vector storage; orchestrated a local Ollama (Llama-3) LLM for grounded Q&A.

Traffic Signal Management GitHub ↗
Stack: Multi-Agent Reinforcement Learning (PPO), SUMO, Python
- Developed an RL model using PPO to optimize traffic signal timings based on SUMO-simulated real-world traffic data.
- Designed reward strategies to prioritize high-traffic lanes and reduce overall urban congestion.

ArcFlow – Interactive Story Platform GitHub ↗
Stack: React, FastAPI, MongoDB, Generative AI API
- Built an AI-driven story platform with dynamic branching choices using React, FastAPI, and MongoDB.
- Integrated a genre/mood-aware generative API for narrative progression and implemented secure per-user story history management.

ACHIEVEMENTS & CERTIFICATIONS

IBM Generative AI Professional Certificate | Solved 200+ problems on LeetCode & GeeksforGeeks